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On the basis of ideal characteristics of an op-amp, following rules can be applied for analysing the operation of any op-amp circuit:

1. **Voltage rule.** The output attempts to do whatever is necessary to make the voltage difference between the inputs zero.

2. **Current rule.** The inputs draw no current.

Applications. Op-amps are the basic building block of analogue computers for performing various mathematical operations like addition, subtraction, integration and differentiation. These are used to build amplifiers, active filters, R-C oscillators, comparators, wave-shaping circuits, differential amplifiers, voltage-to-current

converters (and vice versa), logarithmic amplifiers, analogue multipliers and square-wave generators.

On the basis of voltage and current rules, two applications have been analysed here: inverting amplifier and non-inverting amplifier.

Inverting amplifier. Fig. 3 shows the circuit of an inverting amplifier. The op-amp is connected using resistors R_1 and R_f . The input signal is applied in series with R_1 and the output is connected back to the inverting input through R_f . The non-inverting input is connected to the ground reference. As the input signal becomes positive, the output will turn negative and vice versa. The output voltage depends on the ratio of the two resistors (R_1 and R_f).

Current through $R_1 = V_{IN}/R_1$

Current through $R_f = V_{OUT}/R_f$

Therefore voltage amplification

$V_{OUT}/V_{IN} = -R_f/R_1$

Non-inverting amplifier. The input signal is connected directly to the non-inverting input (+) and input resistor R_1 is grounded as shown in Fig. 4. As

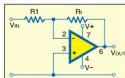


Fig. 3: Inverting amplifier

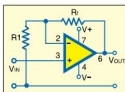


Fig. 4: Non-inverting amplifier

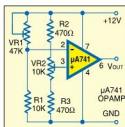


Fig. 5: μ A741-Based operational amplifier circuit

the signal moves in either direction, the output follows in phase to maintain the inverting input at the same voltage.

$V_{gain} = 1 + R_f/R_1$

One of the important applications of op-amp is as comparator. A comparator compares an input voltage with a reference voltage. Basically, it swings its output to + or -Vcc at the slightest difference between its inputs. With one of the inputs as reference voltage, the output of the comparator indicates whether the input voltage (V_{IN}) is above or below the reference voltage (V_{ref}). When V_{IN} is slightly greater than V_{ref} , the output voltage approaches the positive supply voltage. When V_{IN} is slightly less than V_{ref} , the output voltage approaches negative supply voltage.

Fig. 5 shows a practical circuit based on μ A741 op-amp configured in comparator mode. The reference voltage is maintained at the non-inverting input (pin 3) by the voltage divider network comprising R_2 , VR_2 and R_3 . The signal to be compared is applied at the inverting input (pin 2). When the voltage at pin 2 is greater than the reference voltage, the output voltage at pin 6 will be 0V. And if the voltage at pin 2 is less than the reference voltage, the output voltage at pin 6 will be nearly 12 V.

Some popular op-amp devices are TL251, LM339 (dual-comp), μ A741, CA3140 and LM324 (quad amp).

For more applications of op-amps, you may refer to *Chip-Talk Electronic Hobby Projects* book published by EBY.

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